
Multidisciplinary Project Report 1

SnowFLOWS: Generative Spatio-Temporal Modelling of Snow Cover Dynamics Using Satellite Imagery

By:

Carla Youness (232771)
Nour el Saghir (230520)
Reina Taher (230134)
Ali Mansour (231039)
Hassan Dakhllallah (222814)

Supervisor(s):

Dr. Laurent Drapeau
Dr. Youssef El Bakouny

01/03/2026

Ecole Supérieure d'Ingénieurs de Beyrouth
Université Saint-Joseph de Beyrouth

Contents

1	Introduction	3
2	Problem requirements and constraints	5
2.1	Functional requirements	5
2.2	Constraints	6
2.2.1	Technical constraints	6
2.2.2	Non-technical constraints	7
2.2.3	Standards / codes / regulations / policies	8
3	Background / Research / Related work / Literature review	9
4	Solution design	11
4.1	Multiple design alternatives comparison	11
4.2	Required resources	14
4.3	Conceptual model	15
5	Project Management	18
5.1	Planning	18
5.2	Project Facts and Figures	19
5.3	Conflict management	20
6	Conclusion	20
7	References	20

1 Introduction

Snow cover in Lebanese mountainous regions plays a fundamental role in hydrology, water resource management, agriculture, tourism, and seasonal climate regulation. With elevations exceeding 3000 meters, Lebanon's terrain experiences significant seasonal snowfall that directly influences groundwater recharge and river discharge. Recent variability in snowfall patterns increases the need for reliable monitoring and predictive modeling of snow cover dynamics.

Problem identification:

Accurately estimating and forecasting snow distribution over Lebanese mountains is a complex engineering problem. Snow accumulation depends on interacting factors such as altitude, slope orientation, meteorological variability, and surface conditions. Satellite-based snow detection also presents technical challenges, particularly the separation of snow from clouds in optical imagery and the handling of temporal discontinuities due to cloud cover. In addition, large historical archives (approximately 10–15 years) must be processed efficiently and consistently. Google Earth can be used as a baseline visual reference to support area selection and qualitative verification of outputs.

The complexity of the problem arises from:

- Integration of long-term multi-temporal satellite datasets over 10–15 years (Sentinel-2 optical, Sentinel-1 SAR), and associated meteorological/topographic information.
- Fusion of multi-sensor imagery with topographic parameters (DEM/MNT, slope, aspect).
- Structuring satellite time-series as spatio-temporal sequences (video-like evolution).
- Development and comparison of deep learning models for forecasting snow cover evolution.
- Forecasting short- and medium-term snow dynamics (+3, +10, +30 days) with physically plausible behavior.
- Evaluation of prediction performance across altitude ranges, slope orientations, and terrain conditions.
- Production of interpretable outputs (predictive maps, graphs, and animations) and a 3D terrain visualization.
- Involvement of multiple stakeholders (water authorities, environmental agencies, municipalities, and tourism-related actors).

There is no straightforward solution: snow–cloud separation, multi-sensor integration, large dataset sizes, and the requirement for realistic spatio-temporal forecasting introduce significant technical constraints. The problem is multidisciplinary, combining remote sensing, geospatial processing, machine learning, and scientific visualization. Therefore, a structured engineering design is required to build an integrated system for snow detection, forecasting, validation, and visualization.

Project objectives and outcomes:

The primary objective is to design and implement an integrated spatio-temporal processing and forecasting pipeline combining Sentinel-2 optical, Sentinel-1 SAR, topographic parameters, and meteorological data to predict snow cover evolution in Lebanese mountain regions.

The specific objectives are:

1. Build a multi-sensor snow dataset combining Sentinel-2 optical and Sentinel-1 SAR imagery (with Google Earth used as a baseline visual reference) over approximately 10–15 years.
2. Develop a processing pipeline to generate consistent snow information over time, including handling clouds and temporal gaps.
3. Integrate topographic parameters (DEM/MNT, slope, aspect) and meteorological data into the modeling framework.
4. Structure satellite time-series as spatio-temporal sequences to model snow evolution as a dynamic process.
5. Implement and compare deep learning architectures for snow evolution forecasting (ConvLSTM, UNet-3D, TimeSformer, and video diffusion/generative approaches).
6. Produce short- and medium-term forecasts of snow evolution (+3, +10, +30 days).
7. Validate performance using time-based testing (e.g., training on an initial period/week and testing on a subsequent period/week) and evaluate accuracy across altitude and terrain conditions.
8. Generate outputs including graphs of snow-covered surface/quantity evolution, predictive maps, animations, and a 3D mountain visualization with predicted snow overlay.
9. Complete an initial phase of tool and workflow familiarization to ensure reliable implementation (data access, preprocessing, snow detection, visualization).

Expected outcomes include:

- A structured multi-sensor snow time-series dataset and preprocessing scripts.
- Trained AI models for snow-evolution prediction.
- Quantitative evaluation metrics (including terrain-based analysis by altitude/orientation).
- Predictive snow maps, graphs, and video-like animations.
- A prototype visualization tool (web or desktop) and a 3D terrain-based display of predictions.

Report outline:

Section 2 defines the functional requirements and constraints of the system.

Section 3 reviews related work in snow detection, multi-sensor data fusion, and spatio-temporal forecasting.

Section 4 presents the proposed solution design and comparison of modeling alternatives.

Section 5 outlines project management, planning, and risk mitigation.

Section 6 concludes the report.

2 Problem requirements and constraints

2.1 Functional requirements

This section defines the core functionalities that must be achieved in order to consider the problem successfully addressed. The system must implement an integrated spatio-temporal snow monitoring and forecasting framework capable of handling multi-sensor satellite data and producing reliable predictive outputs.

The following main functional requirements are defined:

FR1 – Multi-Sensor Data Acquisition and Processing Pipeline

The system shall:

- Retrieve and process multi-temporal Sentinel-2 optical imagery.
- Retrieve and process Sentinel-1 SAR imagery to complement optical data.
- Integrate Digital Elevation Model (DEM/MNT), slope, and aspect data.
- Incorporate meteorological data (e.g., ERA5 temperature and precipitation).
- Perform cloud masking and preprocessing to ensure temporal consistency.
- Generate consistent snow-coverage maps over a 10–15 year historical period.

This requirement ensures the creation of a structured and reliable snow time-series dataset.

FR2 – Spatio-Temporal Snow Forecasting Model

The system shall:

- Structure multi-temporal satellite data as spatio-temporal sequences.
- Implement deep learning architectures including:
 - ConvLSTM
 - UNet-3D
 - TimeSformer
 - Diffusion-based generative models (if feasible within time constraints)
- Produce short- and medium-term snow forecasts at:
 - +3 days
 - +10 days
 - +30 days
- Ensure physically plausible snow evolution behavior (accumulation/melting patterns consistent with meteorological inputs).

This requirement constitutes the core AI modeling component of the project.

FR3 – Model Evaluation and Terrain-Based Analysis

The system shall:

- Split data temporally into training and testing periods.
- Evaluate model performance using quantitative metrics (RMSE, MAE, IoU, Dice coefficient where applicable).
- Analyze prediction accuracy across:
 - Altitude ranges
 - Slope orientations (aspect)
 - Terrain steepness
- Compare multiple architectures and select the most suitable model.

FR4 – Visualization and Prototype Deployment

The system shall:

- Generate predictive snow maps for each forecast horizon.
- Produce time-series graphs of snow surface evolution.
- Create video-like animations representing snow dynamics.
- Provide a prototype visualization interface (web or desktop-based).
- Integrate a 3D terrain representation with predicted snow overlay.

2.2 Constraints

The system must operate within a set of technical and non-technical constraints defined by the project scope and available resources.

2.2.1 Technical constraints

The following technical constraints apply:

1. Large Dataset Size

- Sentinel-1 and Sentinel-2 archives over 10–15 years generate large data volumes.
- Efficient memory management and cloud-based processing (e.g., Google Earth Engine) are required.

2. Cloud Cover Variability

- Optical imagery may contain significant cloud coverage.
- Cloud masking and multi-sensor fusion (SAR + optical) must mitigate temporal discontinuities.

3. Physical Plausibility Constraint

- Predictions must respect realistic snow accumulation and melting behavior.
- Forecast outputs must remain consistent with meteorological drivers.

4. Computational Resources

- Deep learning model training must be feasible within available GPU or cloud resources.
- Model complexity must be balanced against training time and memory usage.

5. Software Environment

- Implementation shall be conducted using Python-based frameworks (PyTorch or TensorFlow).
- Geospatial processing shall comply with raster data handling libraries (e.g., rasterio, xarray, GEE API).

2.2.2 Non-technical constraints

The following non-technical constraints are considered:

1. Schedule Constraint

- The project must be completed within the academic semester timeline.
- Model selection and experimentation must remain feasible within this timeframe.

2. Usability

- The visualization prototype must present interpretable results for non-technical stakeholders.

3. Extensibility

- The system design should allow future integration of additional meteorological variables or geographic regions.

4. Interoperability

- Output maps must be exportable in standard geospatial formats (GeoTIFF, shapefile, or OGC-compliant formats).

5. Ethical and Environmental Responsibility

- Data usage must comply with open-access satellite data policies.
- No misuse of environmental predictive outputs.

6. Sustainability

- Preference for cloud-based processing to minimize unnecessary local computational energy usage.

2.2.3 Standards / codes / regulations / policies

Although full certification is beyond the scope of this academic project, the system design acknowledges and aligns with relevant standards and policies:

1. Copernicus Sentinel Data Access Standards

- Compliance with official data usage and metadata policies.

2. Google Earth Engine Processing Conventions

- Proper use of platform APIs and geospatial data handling methods.

3. FAIR Data Principles

- Ensuring data is Findable, Accessible, Interoperable, and Reusable.

4. OGC (Open Geospatial Consortium) Standards

- Spatial data representation consistent with standard coordinate reference systems and formats.

5. ISO 19115 (Geographic Information – Metadata)

- Consideration of structured metadata documentation for dataset reproducibility.

These standards guide system reliability, reproducibility, and professional engineering practice.

3 Background / Research / Related work / Literature review

Snow monitoring using satellite imagery has evolved significantly since the launch of early Earth observation missions by the National Aeronautics and Space Administration (NASA). A major advancement came with the Moderate Resolution Imaging Spectroradiometer (MODIS), which enabled daily global snow mapping.

One of the most widely used snow detection techniques is the **Normalized Difference Snow Index (NDSI)**:

$$NDSI = \frac{Green - SWIR}{Green + SWIR}$$

Snow strongly reflects green light and absorbs shortwave infrared radiation, making this index effective for classification. Common datasets used in the literature include Sentinel-2 MSI, Landsat 8 OLI, and MODIS products.

Limitations Identified in Literature

Despite its simplicity and efficiency, NDSI-based detection presents several challenges:

- Snow/cloud misclassification
- Terrain shadow effects in mountainous regions
- Mixed pixels in moderate-resolution imagery
- Threshold sensitivity across different climates

These limitations are particularly relevant in complex terrains such as Lebanon.

Multi-Sensor and Terrain-Aware Approaches

Recent studies integrate optical imagery with radar data from Sentinel-1 to overcome cloud limitations. Additionally, Digital Elevation Models (DEM) are increasingly used to incorporate altitude, slope, and aspect into classification models.

Advantages:

- Improved robustness under cloudy conditions
- Better representation of topographic influence

Drawbacks:

- Higher preprocessing complexity
- Increased computational requirements
- Often limited to classification rather than forecasting

Most existing systems apply terrain features as auxiliary variables rather than embedding them into deep spatio-temporal models.

Snow Forecasting Methods

Snow forecasting approaches in literature can be grouped into three categories:

1. Statistical Models (e.g., ARIMA)

- Lightweight and interpretable
- Limited capacity to model nonlinear spatial dynamics

2. Machine Learning Models (e.g., Random Forest, LSTM)

- Capture nonlinear relationships
- Often treat spatial and temporal components separately

3. Deep Learning Spatio-Temporal Models (e.g., ConvLSTM, 3D CNN)

- Joint spatial-temporal modeling
- Higher predictive accuracy
- Computationally demanding
- Mostly applied to large continental regions

Few studies focus specifically on small mountainous Mediterranean countries.

Identified Gaps and Project Positioning

From the reviewed literature, several gaps emerge:

- Limited snow forecasting systems tailored to small mountainous countries
- Insufficient integration of Mediterranean climate variability
- Weak coupling between snow detection and forecasting in unified systems
- Limited operational visualization for decision-making

The SnowFLOWS project addresses these gaps by:

- Focusing specifically on Lebanon
- Integrating optical, radar, meteorological, and terrain data
- Implementing spatio-temporal deep learning (ConvLSTM)
- Combining classification and forecasting within a single framework
- Providing visualization outputs for practical interpretation

Unlike traditional approaches that stop at detection, SnowFLOWS proposes an integrated pipeline from multi-sensor classification to predictive modeling and visualization.

4 Solution design

4.1 Multiple Design Alternatives Comparison

Designing a snow detection and forecasting system involves several possible architectural and modeling choices. Before selecting the final architecture, multiple alternatives were evaluated against the project requirements:

- Accurate snow classification
- Reliable multi-day forecasting
- Adaptation to Lebanese mountainous terrain
- Moderate computational constraints
- Integration of heterogeneous data sources

Alternative 1: Classical Index-Based + Statistical Forecasting

Architecture

- Snow detection using NDSI thresholding.
- Time-series forecasting using ARIMA on historical snow coverage.

Advantages

- Low computational cost.
- Simple implementation.
- High interpretability.

Limitations

- Sensitive to cloud contamination.
- Poor handling of nonlinear snow dynamics.
- Limited spatial modeling capability.
- Forecasting disconnected from spatial features.

Evaluation

While computationally efficient, this approach does not satisfy the requirement for spatially consistent forecasting across complex terrain.

Alternative 2: Machine Learning-Based Classification + LSTM Forecasting

Architecture

- Random Forest classifier using spectral bands and terrain features.
- LSTM model for temporal prediction of snow coverage.

Advantages

- Captures nonlinear relationships.
- Better generalization than index-based methods.
- Integrates meteorological variables.

Limitations

- Spatial information treated independently from temporal evolution.
- Requires feature engineering.
- Moderate computational complexity.

Evaluation

This approach improves predictive capability but does not fully exploit spatial dependencies within satellite imagery.

Alternative 3 (Selected Design): Multi-Sensor Spatio-Temporal Deep Learning

Architecture

- Multi-sensor fusion (Sentinel-1 SAR + Sentinel-2 optical + DEM + meteorological data).
- Snow classification via convolutional feature extraction.
- Forecasting using ConvLSTM for joint spatial-temporal modeling.
- Unified pipeline for detection and forecasting.

Advantages

- Captures both spatial and temporal dependencies.
- More robust to cloud cover (SAR integration).
- Explicit integration of terrain influence.
- End-to-end learning framework.

Limitations

- Higher computational requirements.
- Requires structured dataset preparation.

Evaluation

This alternative best satisfies functional requirements, especially:

- Spatial coherence in mountainous terrain.
- Forecasting capability beyond simple regression.
- Integration of heterogeneous data.

Therefore, Alternative 3 was selected as the final system design.

High-Level System Design

The proposed system consists of five main modules:

1. Data Acquisition Module

- Sentinel-1 SAR
- Sentinel-2 optical imagery
- Digital Elevation Model (DEM)
- Meteorological datasets

2. Preprocessing Module

- Radiometric correction
- Cloud masking
- Geometric alignment
- Terrain feature extraction

3. Snow Classification Module

- Convolutional feature extraction
- Pixel-level snow detection

4. Forecasting Module

- Spatio-temporal modeling using ConvLSTM
- Multi-day prediction horizons (+3, +10, +30 days)

5. Visualization Module

- Snow maps
- Temporal evolution graphs
- Comparative performance metrics

Low-Level Design Choices

- Input representation: 4D tensors (time $\times$ height $\times$ width $\times$ features).
- Feature channels: spectral bands, SAR backscatter, altitude, slope, temperature, precipitation.
- Loss function: Binary Cross-Entropy for classification; Mean Squared Error for forecasting.
- Evaluation metrics:
 - Accuracy
 - F1-score
 - RMSE for prediction
- Training strategy: Sliding temporal window approach.

Problem formulation

The project addresses the problem of snow cover detection and forecasting over Lebanon using multi-sensor satellite data.

Let:

- X_t = multi-sensor satellite tensor at time t
- T = terrain features (altitude, slope, aspect)
- M_t = meteorological variables at time t
- Y_t = binary snow cover map at time t

Snow Detection Problem

Determine a mapping:

$$f: (X_t, T, M_t) \rightarrow Y_t$$

where f classifies each pixel as snow or non-snow.

Snow Forecasting Problem

Given a historical sequence:

$$\{Y_{t-n}, \dots, Y_t\}$$

Predict future snow distribution:

$$\hat{Y}_{t+k} = g(Y_{t-n:t}, T, M_{t:t+k})$$

where:

- g is a spatio-temporal model (ConvLSTM),
- $k \in \{3, 10, 30\}$ days.

Engineering Constraints

- Limited computational resources (GPU/CPU constraints).
- Incomplete data due to cloud coverage.
- Terrain-induced reflectance variability.
- Moderate dataset size.

Assumptions

- Satellite data are properly co-registered.
- DEM accurately represents topographic features.
- Meteorological forecasts are available for future horizons.
- Snow classification ground truth is sufficiently reliable.

4.2 Required resources

The SnowFLOWS project requires a combination of software, data, and hardware resources to handle multi-sensor satellite imagery, perform preprocessing, train spatio-temporal models, and generate visual outputs.

Software:

- Python 3.x with libraries for numerical processing, satellite data handling, and deep learning (NumPy, Pandas, Rasterio, OpenCV, PyTorch/TensorFlow)
- Geospatial libraries for handling spatial data (GDAL, Shapely, GeoPandas)
- Visualization tools for graphs, animations, and 3D rendering (Matplotlib, Plotly)
- Google Earth Engine (cloud-based processing of Sentinel archives)
- Optional GIS software (QGIS or ArcGIS) for spatial inspection

Data:

- Historical multi-sensor satellite imagery (Sentinel-1 SAR, Sentinel-2 optical) covering 10–15 years
- Terrain data for mountainous reconstruction (DEM, slope, aspect)
- ERA5 meteorological reanalysis data (temperature, precipitation, snow depth)

Hardware:

- Personal computer or workstation with sufficient processing capacity, ideally including:
 - A workstation with 8–16 GB RAM and optional GPU acceleration is recommended for deep learning model training.
 - Stable internet connection for data acquisition

4.3 Conceptual model

The SnowFLOWS conceptual model represents the end-to-end workflow of the system and illustrates the interaction between data, processing, modeling, and visualization components.

System Workflow:

Data Acquisition → Preprocessing → Dataset Construction → AI Modeling → Forecasting → Evaluation & Visualization

1. Data Acquisition Layer

The system integrates heterogeneous geospatial and meteorological datasets, including:

- Sentinel-2 optical imagery (multispectral reflectance)
- Sentinel-1 SAR imagery (backscatter information)
- Digital Elevation Model (DEM) and derived slope data
- ERA5 meteorological data (temperature and precipitation)

2. Preprocessing Layer

The preprocessing stage ensures temporal and spatial consistency of multi-sensor inputs. It includes:

- Cloud masking for optical imagery
- Co-registration between Sentinel-1 and Sentinel-2 datasets
- Feature extraction (e.g., spectral indices, SAR backscatter features)
- Terrain feature derivation (slope computation from DEM)

3. Dataset Construction Layer

Processed data are structured into spatio-temporal sequences representing the evolution of snow cover over time.

Each sample is formatted as a temporal tensor suitable for deep learning models. This stage transforms raw geospatial rasters into organized time-series data cubes.

4. AI Modeling Layer

The system implements and compares multiple deep learning architectures for snow evolution forecasting, including:

- ConvLSTM
- UNet-3D
- TimeSformer

These architectures model snow dynamics as a video-like process, learning spatial-temporal dependencies from historical sequences.

5. Forecasting Layer

The trained models generate short- and medium-term forecasts of snow cover evolution at:

- +3 days
- +10 days
- +30 days

Predictions are constrained to remain physically plausible with respect to seasonal accumulation and melting behavior.

6. Evaluation and Visualization Layer

The final stage includes:

- Quantitative accuracy assessment (RMSE, MAE, IoU)
- Terrain-based evaluation across altitude and slope conditions
- Generation of predictive maps
- Time-series graphs of snow surface evolution
- Video-like animations of snow dynamics
- 3D terrain visualization with predicted snow overlay

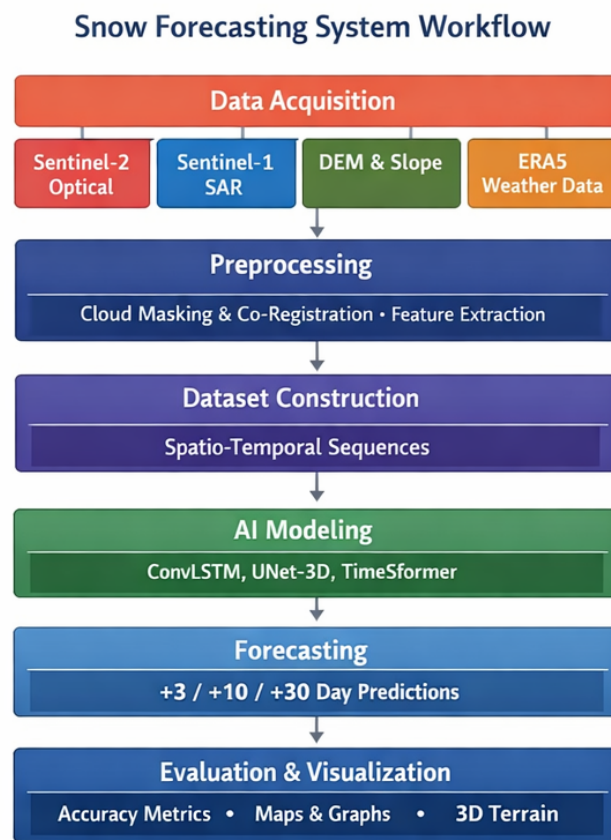


Figure 1: Solution diagram

5 Project Management

5.1 Planning

The project is structured over 10 weeks (2.5 months), following a phased methodology that incorporates simulation, prototype development, and iterative improvement.

Project Phases:

Phase 1 – Tool Familiarization (Weeks 1–2)

- Exploration of satellite data formats (Sentinel-1, Sentinel-2)
- Testing Google Earth Engine workflows
- Preliminary experimentation with snow detection indices
- Definition of study region boundaries

Phase 2 – Data Collection and Preprocessing (Weeks 2–4)

- Retrieval of historical multi-sensor imagery
- Cloud masking and co-registration
- DEM integration and slope extraction
- Spatial standardization and dataset preparation

Phase 3 – Snow Detection and Dataset Construction (Weeks 4–6)

- Generation of snow masks for all timestamps
- Aggregation of snow coverage metrics
- Construction of spatio-temporal tensors for modeling

Phase 4 – Model Development and Training (Weeks 6–8)

- Implementation of ConvLSTM baseline model
- Training on historical sequences
- Hyperparameter tuning
- Optional testing of alternative architectures (UNet-3D, TimeSformer)

Phase 5 – Validation and Testing (Week 9)

- Time-based data splitting (training vs testing period)
- Quantitative evaluation using selected performance metrics
- Qualitative verification using visual comparison tools

Phase 6 – Visualization and Prototype Integration (Weeks 9–10)

- Generation of predictive snow maps
- Time-series graph creation
- Animation production
- 3D terrain visualization with snow overlay

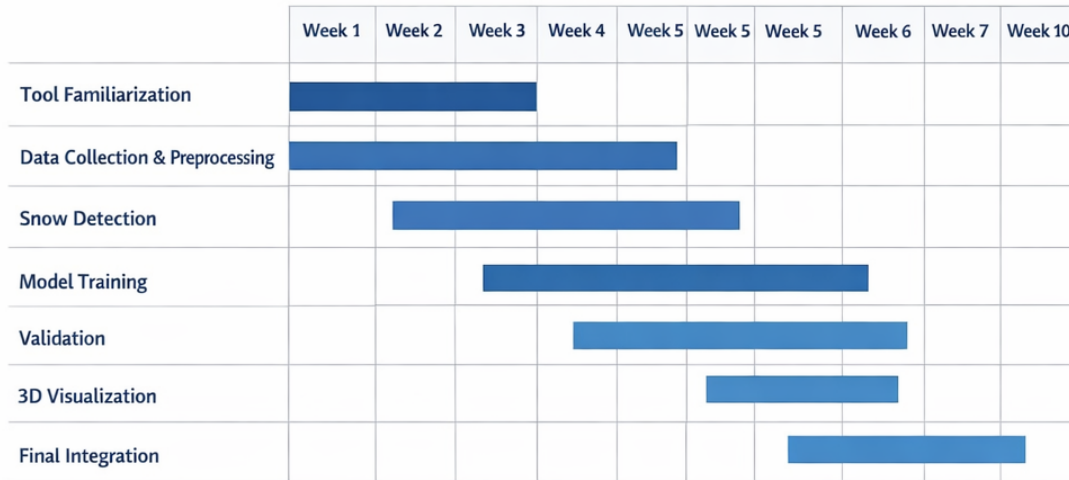


Figure 2: Project Gantt Chart

5.2 Project Facts and Figures

Collaborative Work Structure

All members will:

- Participate in dataset construction.
- Contribute to model implementation and testing.
- Participate in evaluation experiments.
- Contribute to visualization development.
- Rotate documentation and report writing responsibilities.
- Attend weekly technical integration meetings.

Team Facts & Roles (5 members - all CCE)

Name	Affiliation	Lead Responsibility	Supporting Responsibilities
Hassan Dakhallallah	CCE	Lead: Sentinel-2 preprocessing and optical snow extraction	Support in SAR integration, modeling experiments, and evaluation
Carla Youness	CCE	Lead: Sentinel-1 SAR preprocessing and multi-sensor alignment	Support in snow detection, terrain integration, and validation
Reina Taher	CCE	Lead: Terrain & meteorological feature integration (DEM, slope, aspect, ERA5)	Support in dataset preparation and model training
Nour Saghir	CCE	Lead: Deep learning modeling (ConvLSTM, UNet-3D, TimeSformer)	Support in preprocessing, evaluation, and visualization
Ali Mansour	CCE	Lead: Evaluation, forecasting analysis, and visualization prototype (maps, animations, 3D view)	Support in dataset structuring and model testing

5.3 Conflict management

To ensure smooth collaboration, the team adopts the following conflict management strategy.

1) A team member is late delivering their part

- Work is organized into weekly milestones aligned with the integration meetings.
- If a delay exceeds 48-72 hours, the team triggers a recovery plan: redistribute subtasks temporarily, pair-work to unblock technical issues, and adjust scope of non-critical features while preserving core deliverables.
- Each milestone ends with a short “done” check (outputs + code/results pushed + short note).

2) A dependency/provider is late (data access, software, compute resources)

- Maintain a Plan B: alternative dataset source, reduced-resolution experiments, or simplified baseline model.
- Start with an MVP pipeline (minimal working preprocessing → baseline model → basic evaluation), then iterate.
- Use downtime to progress on parallel tasks (documentation, evaluation protocol, visualization mockups).

3) Conflict between members (technical disagreement or coordination)

- Decision rule: define requirement, compare 2-3 options, and decide using criteria (accuracy, feasibility, time, complexity).
- Final decision is documented (short paragraph) to avoid repeating the same debate.
- If unresolved, escalate to the supervisor for arbitration.
- Documentation/report writing is rotated, reducing workload imbalance (a common source of conflict).

6 Conclusion

This proposal presented The SnowFLOWS project and defined a clear collaborative structure for building an end-to-end snow monitoring and forecasting pipeline. The team responsibilities were distributed across key technical blocks: Sentinel-2 optical snow extraction, Sentinel-1 SAR preprocessing and alignment, terrain/meteorological feature integration, deep learning modeling, and evaluation/visualization. A conflict management strategy was also established to mitigate delays, dependency risks, and technical disagreements. The next phase will focus on implementing the full pipeline, running experiments and evaluations, and delivering validated results along with clear visual outputs and documentation.

7 References

- [1] European Space Agency (ESA). Sentinel-1 Mission Overview and SAR Products Documentation, ESA/Copernicus documentation.
- [2] European Space Agency (ESA). Sentinel-2 Mission Overview and MSI Products Documentation, ESA/Copernicus documentation.
- [3] Hersbach, H. et al. The ERA5 global reanalysis, ECMWF / Copernicus Climate Change Service documentation.
- [4] Ronneberger, O., Fischer, P., Brox, T. "U-Net: Convolutional Networks for Biomedical Image Segmentation", MICCAI, 2015.
- [5] Shi, X. et al. "Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting", NeurIPS, 2015.
- [6] Bertasius, G., Wang, H., Torresani, L. "Is Space-Time Attention All You Need for Video Understanding?", ICML, 2021.
- [7] Paszke, A. et al. "PyTorch: An Imperative Style, High-Performance Deep Learning Library", NeurIPS, 2019.